AWS Route 53 vs Self-Hosted DNS Server

1. Introduction

DNS (Domain Name System) is a fundamental component of the internet that translates human-readable domain names into IP addresses.

In this project, DNS is implemented in two ways:

- **Self-hosted DNS server (BIND9/CoreDNS on EC2)**
- **AWS Route 53 (Managed DNS service)**

The goal is to understand how DNS resolution works and how switching **NS (Nameserver) records** changes the authoritative DNS system.

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2. What is DNS? (Core Concept)

2.1 Definition

DNS is a hierarchical system that resolves:

```
example.com → 192.168.1.1
```

2.2 DNS Resolution Flow

```
User → Resolver → Root Server → TLD Server → Authoritative DNS → IP Address
```

Key Terms

Term	Meaning
Resolver	Client making DNS request
Authoritative Server	Final source of truth
Zone	DNS records for a domain
NS Record	Points to authoritative server
TTL	Cache duration

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3. Self-Hosted DNS Server

🔹 3.1 What is Self-Hosted DNS?

A DNS server hosted manually on a machine (EC2), using tools like:

- BIND9
- CoreDNS

3.2 Components

Component	Role
BIND9/CoreDNS	DNS server software
Zone File	Stores DNS records
EC2 Instance	Hosting server
Security Group	Controls access

3.3 DNS Records (Example)

```
A record: example.com → 1.2.3.4
CNAME: www → example.com
```

3.4 How It Works

```
User → Domain Registrar → Self-hosted DNS → IP Response
```

3.5 Security Considerations

- Open port 53 (TCP/UDP)
- Prevent DNS amplification attacks
- Configure firewall rules

3.6 Challenges

- Manual setup
- Maintenance required
- No automatic scaling

- High availability must be configured manually

3.7 Use in Your Task

- Hosted on EC2
- Used as **authoritative DNS** when NS points to it
- Tested using `dig` and `nslookup`

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4. AWS Route 53

🔹 4.1 What is Route 53?

Route 53 is a **fully managed DNS service** provided by AWS.

4.2 Key Features

- Highly available
- Globally distributed
- Managed service
- Supports advanced routing

4.3 How It Works

```
User → Route 53 → DNS Record → IP Address
```

4.4 Hosted Zone

A hosted zone is a container for DNS records.

Types:

- Public Hosted Zone
- Private Hosted Zone

4.5 Supported Record Types

- A (IP mapping)
- CNAME

- MX (mail)
- TXT
- NS

4.6 Routing Policies

Route 53 supports:

- Simple routing
- Weighted routing
- Latency-based routing
- Failover routing

4.7 Security Features

- DDoS protection (via AWS Shield)
- IAM integration
- Health checks

4.8 Advantages

- No server management
- Automatic scaling
- High reliability

4.9 Use in Your Task

- Hosted zone created for domain
- DNS records added
- Used after switching NS records

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5. NS Record (Critical Concept)

🔹 5.1 What is NS Record?

NS record defines which DNS server is **authoritative** for a domain.

5.2 Role in Your Task

Switching NS = Switching DNS system

```
NS → Self-hosted DNS → Uses EC2 DNS
NS → Route 53 → Uses AWS DNS
```

Important

- Only one system is active at a time
- Depends on domain registrar configuration

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6. DNS Switching Process (Your Task Core)

🔹 **Step 1: Use Self-Hosted DNS**

- Update NS to EC2 DNS
- Test using:

```
dig example.com
nslookup example.com
```

🔹 **Step 2: Switch to Route 53**

- Update NS to Route 53 nameservers
- Test again

Observation

- Response source changes
- IP may change
- Authority section changes

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7. DNS Propagation

🔹 **7.1 What is Propagation?**

Time taken for DNS changes to reflect globally.

Factors:

- TTL value
- ISP caching
- Resolver behavior

Typical Time:

- Few minutes to 48 hours

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8. DNS Testing Tools

🔹 8.1 dig

```
dig example.com
```

Shows:

- Answer
- Authority
- Nameserver

🔹 8.2 nslookup

```
nslookup example.com
```

What to Verify

- Which server responded
- Correct IP returned
- CNAME resolution

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9. Route 53 vs Self-Hosted DNS (Comparison)

Comparison Table

Feature	Self-Hosted DNS	Route 53
Type	Manual	Managed

Setup	Complex	Easy
Maintenance	Required	None
Scalability	Limited	High
Availability	Manual setup	Built-in
Cost	Low infra cost	Pay-as-you-go
Security	Manual	AWS-managed
Performance	Depends on server	Global edge network

Simple Explanation

👉 Self-hosted DNS = Control + complexity
👉 Route 53 = Simplicity + scalability

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10. Why This Task is Important

Learning Outcomes

- Understand DNS resolution
- Learn authoritative DNS switching
- Compare manual vs managed DNS
- Observe real-world behavior

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11. Real-World Use Cases

- Enterprises migrating to cloud DNS
- Hybrid DNS setups
- Disaster recovery
- Multi-region architectures

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12. Conclusion

- Self-hosted DNS provides flexibility and control
- Route 53 provides scalability, reliability, and ease of use
- NS record switching determines which DNS system is active
- DNS propagation plays a key role in real-world systems

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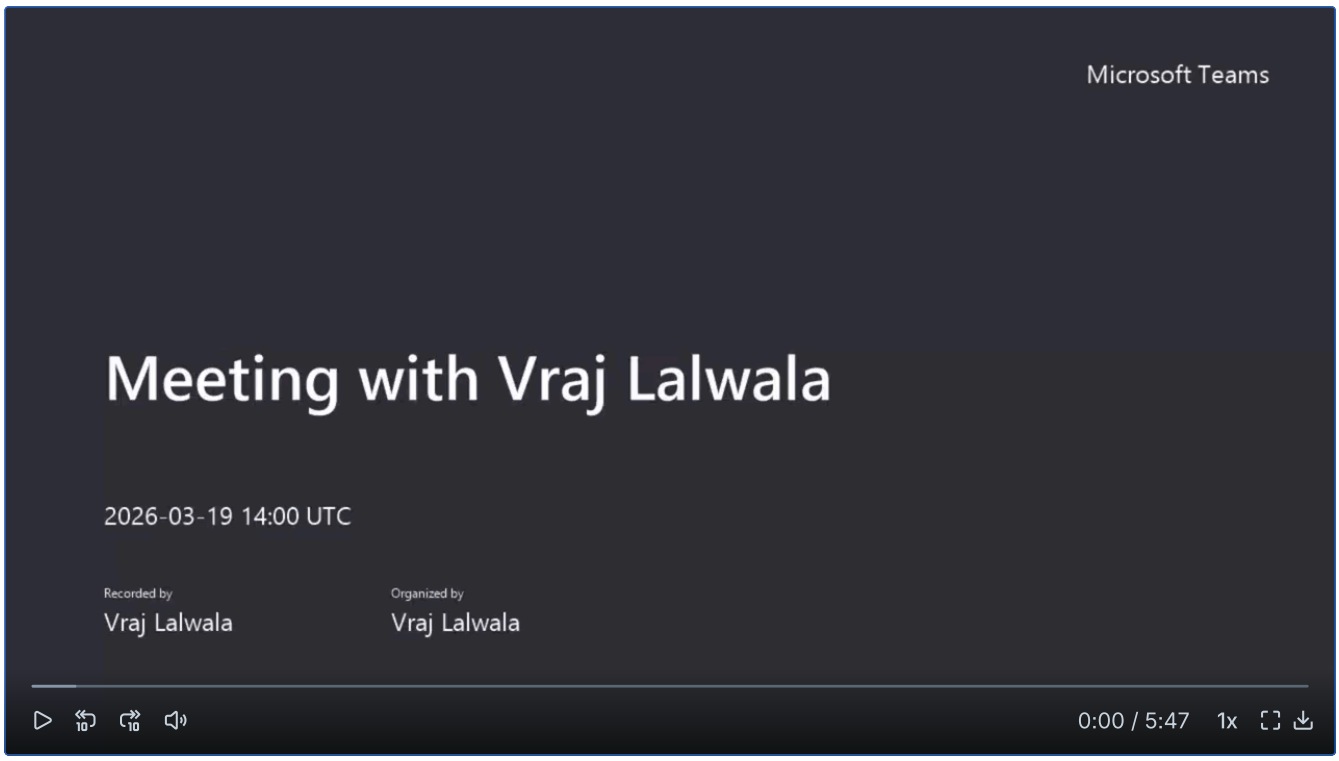
13. Video Explanation